

A Count Without Numbers

The $\sqrt{2}$ spiral, its folded dimension, and why it is not Fibonacci

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Abstract

We describe an elementary construction of a square-generation spiral in which each generation is obtained from the previous one by scaling by $\sqrt{2}$ and turning through 45° . The construction is deliberately **numberless**: it is generated as a single continuous directional stroke – three straight steps and one diagonal, alternating – whose lengths are fixed not by measurement but by one rule, that each cycle returns to a shared origin. The generator is the Gaussian integer $1 + i$, so the figure is the orbit of the unit square under $z \mapsto (1 \pm i)z$, about a pole it never occupies. From this single object we read three things: a **folded (transitional) dimension** – the 45° -offset generations, which are exactly the odd powers of $1 + i$; an **invariance under the shape of the unit**, so the same spiral may be drawn with square, oval, diamond, rectangular or curved units without change; and a clean **separation from both the golden (logarithmic) spiral and the Fibonacci square construction** with which it is easily confused – from the first by growth rate, from the second by mechanism and range. None of the underlying geometry is new; the contribution is a unified and elementary way of seeing it, in which $\sqrt{2}$ functions less as a number used by the construction than as the language the construction is written in.

1. Introduction

Consider a unit square, and the operation that takes its diagonal to be the side of a next, larger square. Iterated, this produces a nested sequence of squares whose sides run $1, \sqrt{2}, 2, 2\sqrt{2}, 4, \dots$ – each $\sqrt{2}$ times the last – and whose orientations alternate by 45° . The construction is old. That the diagonal square has twice the area is the doubling of *Meno* (Plato, ~385 BCE); the alternating $\sqrt{2}$ -scaled nest is Hambidge's dynamic rectangle and the $\sqrt{2}$ (ISO A-series) rectangle; the multiply-by- $\sqrt{2}$ -and-rotate- 45° step is multiplication by the Gaussian integer $1 + i$; and the curve through the corners is a logarithmic (equiangular) spiral, a spiral similarity in Coxeter's sense. We claim none of this.

What we offer is a way of seeing the same object *whole*, and elementary throughout:

1. that it can be generated as a **single directional stroke with no numbers and no coordinate frame** (§2), the lengths being consequences of one closure rule rather than inputs;

2. that its 45° -offset generations form a **folded, transitional dimension** – precisely the odd powers of $1 + i$ (§4);
3. that the spiral is **invariant under the shape of the unit** placed on it (§5); and
4. that it is **neither** the golden logarithmic spiral **nor** the Fibonacci square construction it may be mistaken for, and that both differences can be stated exactly (§6).

The unifying observation (§7) is that every quantity in the figure – position, scale, rotation, and even the “rectangular” variants – is an expression of $\sqrt{2}$ alone. In that sense $\sqrt{2}$ is not a constant the construction consumes; it is the alphabet it is written in.

2. A count without numbers

The figure can be drawn by hand in one continuous stroke, without lifting the pen, without measuring, and without axes. The instruction is purely directional. Beginning from a point, take:

three straight steps – right, down, left – then one diagonal step, up-right; then three diagonal steps – down-right, down-left, up-left – then one straight step, right;

and repeat, alternating. Each group of four is one generation. The roles of “straight” and “diagonal” exchange every generation: what is a side in one generation is a diagonal in the next. This is the sense in which a diagonal edge “is” a straight edge, depending on the generation it belongs to.

Two features deserve emphasis. First, **no magnitude appears in the instruction** – only directions, under one closure rule: each cycle returns the stroke to the shared origin, closing a square, and the diagonal step that follows is that square’s diagonal, which becomes the first side of the next cycle. The lengths are thereby *determined, not chosen*: a step of arbitrary length would not bring the cycle back to zero, so the closure fixes every length. In particular $\sqrt{2}$ is never named – it is simply the diagonal of a unit square, appearing (by Pythagoras) the moment a square closes, not supplied as an input. Second, the drawing is **one continuous line, but not self-avoiding**: from the third generation onward it rides back over its own earlier segments, always along the axis rays through zero. (The mechanism is that generation $n + 2$ leaves zero along the same ray on which generation n ’s closing side arrived, so each generation from the third traces over the previous-but-one’s closing side for that side’s full length; a second kind of overlap – a generation covering the body of the one eight steps back – begins at generation 9.)

Written out, the ordered sequence of edges – the *count* – reads

			1
1	1	1	$\sqrt{2}$
$\sqrt{2}$	$\sqrt{2}$	$\sqrt{2}$	2
2	2	2	$2\sqrt{2}$
$2\sqrt{2}$	$2\sqrt{2}$	$2\sqrt{2}$	4
...			

Each length occurs four times, as the four sides of its square; the rows group them by the *motion* (three sides of the current square, then the diagonal that launches the next), which is why each row reads as three repeats and one new value. The lone opening 1 is the seed, with the origin itself uncounted. Equivalently the count may be read off the diagonals rather than the sides, giving the same ladder shifted by one rung: $1, \sqrt{2}, 2, 2\sqrt{2}, 4, \dots$

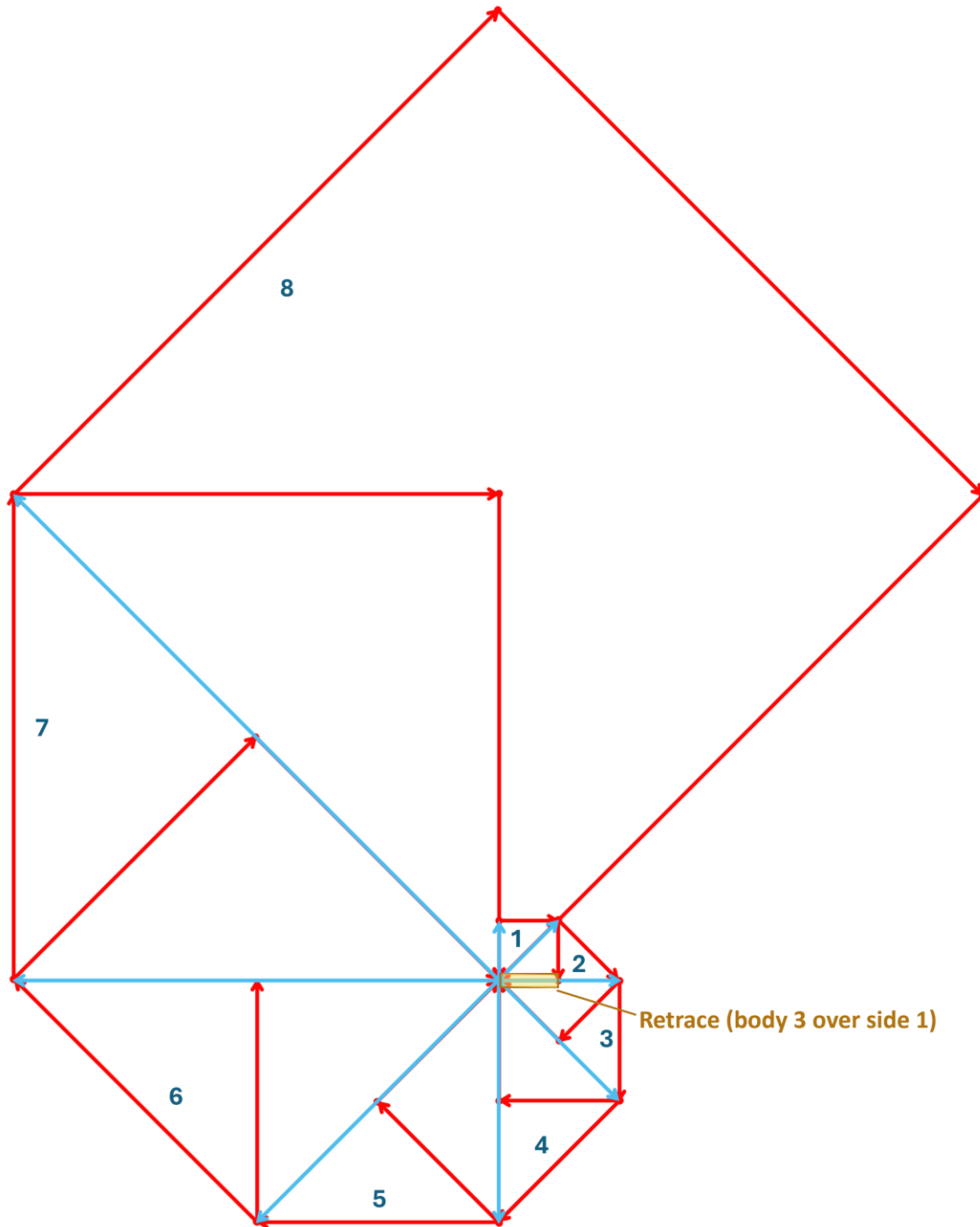


Figure 1. The single stroke through eight generations, one continuous line drawn by direction alone. Blue: the body that launches each generation (the previous generation's diagonal); red: the three sides

that return it to zero. Generations numbered 1–8. Body 3 runs along generation 1's closing side – the first of the retraces described in the text (marked).

3. The generator: $1 + i$

Place the figure in the complex plane with the shared corner at the origin. Multiplication by $1 + i$ scales by $\sqrt{2}$ and rotates by 45° :

$$z \mapsto (1 + i)z; |1 + i| = \sqrt{2}; \arg(1 + i) = 45^\circ.$$

Every generation is therefore the previous one carried by this map (by $1 - i$, if the stroke turns clockwise, as ours does): the figure is the **orbit of the unit square under $z \mapsto (1 \pm i)z$** , each vertex – indeed each point – travelling its own logarithmic spiral

$$z_n = z_0 \cdot (1 \pm i)^n.$$

The origin is the fixed point of this map: a pole that no generation ever occupies. The stroke turns about zero without landing on it – it can circle the origin but never reach it, the way a line tethered to a post can turn about the post but not touch it. This absent, uncounted centre is shared by every generation as a common vertex; the spiral grows *around* it, not away from it, surrounding it after a full turn rather than drifting off in any direction.

4. The folded (transitional) dimension

Because each step advances the orientation by 45° , eight steps advance it by 360° : generation $n + 8$ has the same orientation as generation n , scaled by

$$(\sqrt{2})^8 = 16.$$

Thus generation 1 and generation 9 sit at the same position and rotation, sixteen times the size; the orientations run through eight values and return.

These eight orientations split by parity, and the split is exactly the parity of the powers of $1 + i$. (A note on indexing: generation n is the power $(1 \pm i)^{n-1}$, so the odd-numbered generations are the *even* powers and vice versa; below we index by generation.)

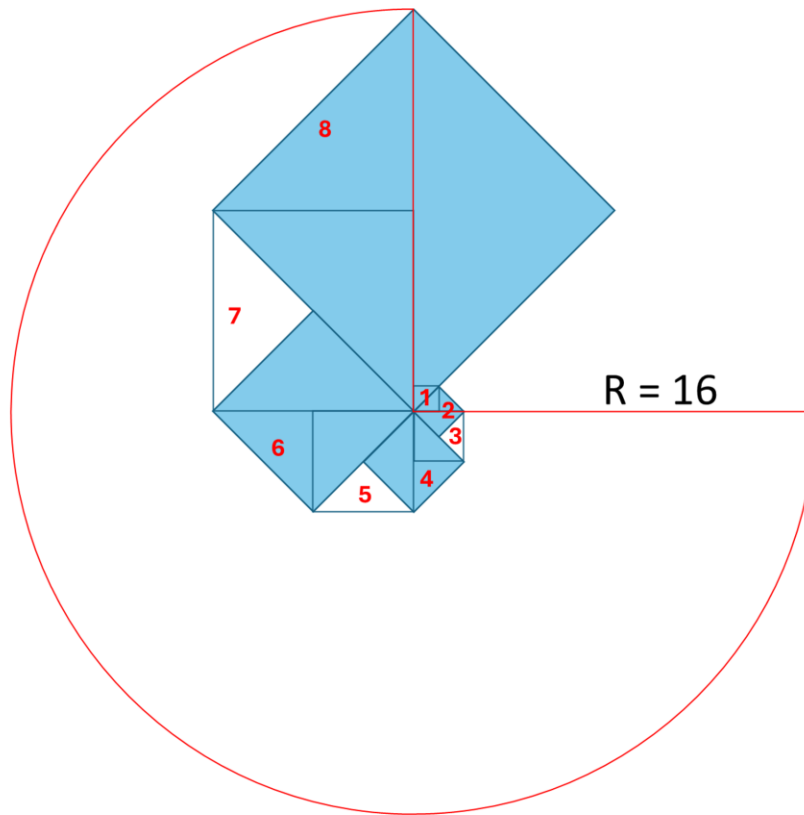
- **On-axis** generations (odd n): orientations $0^\circ, 90^\circ, 180^\circ, 270^\circ$; integer side-lengths 1, 2, 4, 8.
- **Off-axis** generations (even n): orientations $45^\circ, 135^\circ, 225^\circ, 315^\circ$; side-lengths $\sqrt{2}, 2\sqrt{2}, 4\sqrt{2}, 8\sqrt{2}$ – the $\sqrt{2}$ -multiples.

We call the off-axis, 45° -shifted layer the **folded or transitional** dimension: an in-between set of positions through which the construction passes between axis states. It is reached from any on-axis unit by its midpoint-inscribed square – the square on the side-midpoints, half the area and turned 45° – and returned from by that step's inverse, the diagonal. The two parities are two interleaved series – a *normal* series (the integers) and a *folded* series (the $\sqrt{2}$ -multiples) – which together give the side-length ladder

$$(\sqrt{2}/2) - (1) - (\sqrt{2}) - (2) - (2\sqrt{2}) - (4) - (4\sqrt{2}) - (8).$$

The naming $(1/2) - (\sqrt{2}/2)$ is internally exact: it names generation 0 as a **(radicand, root)** pair – area $1/2$ and side $\sqrt{(1/2)} = \sqrt{2}/2$ – mirroring $(1, \sqrt{1}) = (1, 1)$ at generation 1 and $(2, \sqrt{2})$ at generation 2. The count is thus named by its (doubling, root) pair: the radicand doubles (... , $1/2, 1, 2, 4, \dots$) and the side is its root – the same radicand-and-root relation that runs through the whole construction (§2–§3). Generation 0's square is congruent to the *ad quadratum* inner square (the square inscribed on a unit square's midpoints), here positioned with its own corner at zero; and its side $\sqrt{2}/2$ is at once that root, the half-diagonal of the unit square, and $\cos 45^\circ$: one fact wearing several names.

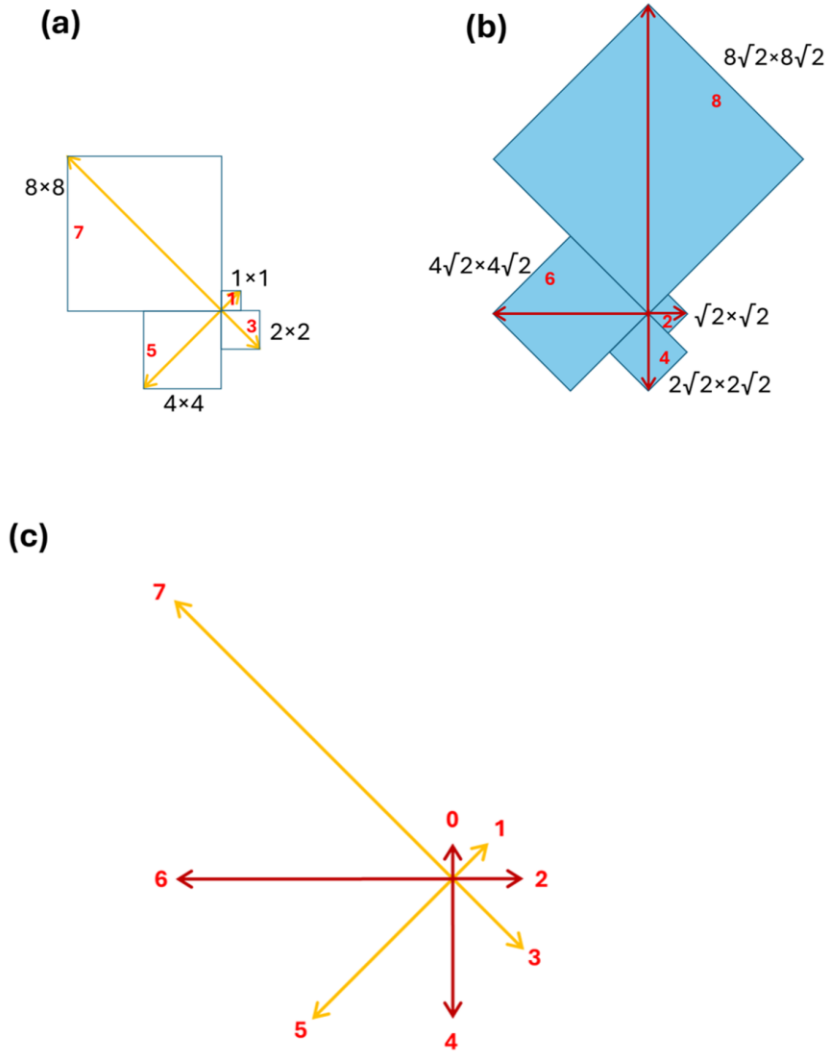
Which parity a chosen unit lands on is a matter of naming, not of geometry: any unit may be declared the 1×1 at 0° , whereupon its neighbours are forced to $\pm 45^\circ$. (The convention that “on-axis” carries integer lengths is only the choice to start from a square aligned with the page.)



A count based on the units' side lengths –

$$1 (1 \times 1) - 2 (\sqrt{2} \times \sqrt{2}) - 3 (2 \times 2) - 4 (2\sqrt{2} \times 2\sqrt{2}) - 5 (4 \times 4) - 6 (4\sqrt{2} \times 4\sqrt{2}) - 7 (8 \times 8) - 8 (8\sqrt{2} \times 8\sqrt{2})$$

Figure 2. The spiral, generations 1–8, inside the reference circle $R = 16$ (the diagonal of generation 8); shaded, the even, folded generations.



A count based on the units' diagonal lengths –
 0 (1) – 1 ($\sqrt{2}$) – 2 (2) – 3 ($2\sqrt{2}$) – 4 (4) – 5 ($4\sqrt{2}$) – 6 (8) – 7 ($8\sqrt{2}$)

Figure 3. The count read off the units. **(a)** the odd (upright) generations, labelled by side-length; **(b)** the even (45°-turned) generations, likewise; **(c)** the eight orientations as a rose, indexed 0–7 by the diagonal count – arrow 0 is the seed direction, and arrow 8 would return to it at length 16. Side-length count: 1 (1x1) – 2 ($\sqrt{2} \times \sqrt{2}$) – 3 (2x2) – 4 ($2\sqrt{2} \times 2\sqrt{2}$) – 5 (4x4) – 6 ($4\sqrt{2} \times 4\sqrt{2}$) – 7 (8x8) – 8 ($8\sqrt{2} \times 8\sqrt{2}$). Diagonal count: 0 (1) – 1 ($\sqrt{2}$) – 2 (2) – 3 ($2\sqrt{2}$) – 4 (4) – 5 ($4\sqrt{2}$) – 6 (8) – 7 ($8\sqrt{2}$).

5. One spiral, any shape

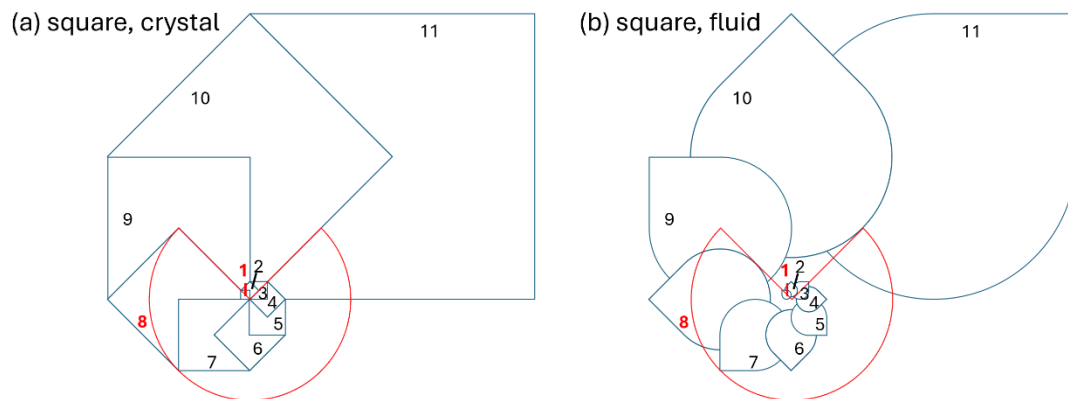
The map $z \mapsto (1 \pm i)z$ is a similarity; every vertex of the figure lies on the logarithmic spiral

$$r(\theta) = r_0 \cdot 2^{(\theta/90^\circ)},$$

whose growth-per-angle is fixed by $|1 + i| = \sqrt{2}$ and $\arg(1 + i) = 45^\circ$, and by nothing else. Consequently **any fixed decoration, placed the same way in each generation's own coordinates and carried rigidly by the similarity, lies on the same family of spirals** – the drawn unit may be anything without disturbing them. This gives freedom in how the unit is drawn:

- **Rendering** – the unit may be *crystal* (rigid and angular: a square, a diamond, a rectangle) or *fluid* (soft and curved: an oval, an arc, a shell). The fluid rendering is the exact analogue of drawing quarter-circles inside Fibonacci squares to obtain the familiar “golden spiral” picture – except the frames here are $\sqrt{2}$ -scaled and 45° -turned.
- **Bounding** – whether the units *bound* one another (their edges meet) or sit *aerated* (spaced, with gaps) is not a free axis but a consequence of the glyph: whether the drawn unit reaches its neighbours. Square units bound in crystal form (their edges meet) but aerate in curved form (inscribed circles leave gaps); the $1:\sqrt{2}$ rectangular units are *observed* to bound in **both** their crystal and curved forms (the extended side reaches the neighbour the inscribed oval would miss). (This is distinct from the deeper overlap structure of §4 – adjacent same-orientation generations, two steps apart, meeting along a ray; the half-area overlap seven steps apart; containment eight steps apart.)

The invariance holds under **one condition**: the unit's shape must not be permitted to redefine the diagonal. If a non-square rectangle is used as a *true* unit, with its own diagonal driving the next side, then the turn is $\arctan(a/b)$ rather than 45° and the growth is $\sqrt{1 + (a/b)^2}$ rather than $\sqrt{2}$; the eight-fold return and the on/off-axis parity are then lost, since closure requires $360^\circ/\theta$ to be an integer. The construction's two-state folded dimension is, in this sense, a square-only structure. The rectangular variants that *preserve* it are $1:\sqrt{2}$ rectangles placed on the *same* $\sqrt{2}/45^\circ$ frame – glyphs, not true units – obtained by scaling one side of a square by $\sqrt{2}$. Under this reading the unit shape is free and the rhythm is kept; under the true-unit reading it is a different spiral.



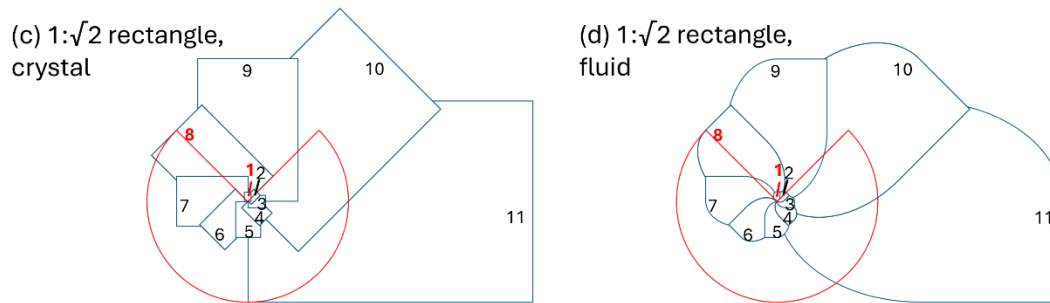


Figure 4. The same eleven-generation spiral rendered with four units: **(a)** square, crystal; **(b)** square, fluid; **(c)** $1:\sqrt{2}$ rectangle, crystal; **(d)** $1:\sqrt{2}$ rectangle, fluid. Generations numbered 1–11; the red arc is a reference circle of radius $8\sqrt{2}$ – the diagonal of generation 7 – with its bounding radii, kept for orientation. Square units bound in (a) and aerate in (b); the rectangular units bound in both (c) and (d).

6. Not Fibonacci

The $\sqrt{2}$ spiral is easily confused with two different things: the **golden logarithmic spiral** of constant ratio ϕ , and the discrete **Fibonacci square construction** that approximates it. It is distinct from each, for different reasons.

- **From the golden spiral – growth.** Both are logarithmic spirals (similarities), and differ only in rate. Ours grows by $\sqrt{2}$ per 45° , i.e. $\times 2$ per quarter turn and $\times 16$ per full turn; the golden spiral grows by $\phi \approx 1.618$ per quarter turn, $\times 6.85$ per turn. They therefore have a different constant tangent-to-radius angle (ours $\approx 66^\circ$, golden $\approx 73^\circ$)¹: different curves, not one curve at two scales. (The nautilus, often invoked alongside, is a third thing again – roughly $\times 3$ per turn, and not golden despite the common claim; Sharp (2002).)
- **From the Fibonacci construction – mechanism.** The Fibonacci construction is *additive* – each square’s side is the sum of the previous two, turning 90° and tiling a rectangle – and its square-vertices only *approach* a logarithmic spiral (their side-ratios tend to ϕ). Ours is *multiplicative* with a constant ratio from the first step, so its vertices lie exactly on a logarithmic spiral from the outset. (The quarter-circle arc usually drawn through Fibonacci squares is itself only an approximation – as, equally, are the fluid glyphs here; the exactness claimed is of the *vertices*, not of any drawn curve.)
- **From the Fibonacci construction – range and scale.** Ours is a *similarity*: any generation, taken as the unit, reproduces the whole series, so there is no

¹ For a logarithmic spiral $r = r_0 \cdot e^{b\theta}$, the constant angle ψ between the tangent and the radius satisfies $\tan \psi = 1/b$, where b is the growth per radian: $b = \ln \rho / (\pi/2)$, with ρ the growth per quarter-turn. For the $\sqrt{2}$ spiral $\rho = 2$ (it doubles every quarter turn), so $b = 2 \ln 2 / \pi \approx 0.441$ and $\psi = \arctan(1/b) \approx 66.2^\circ$; for the golden spiral $\rho = \phi$, $b \approx 0.306$ and $\psi \approx 73.0^\circ$.

privileged starting unit, and the figure runs without end in both directions – outward by the diagonal, inward by the midpoint-inscribed square (the same step reversed, half the area, turned 45°), approaching the origin without reaching it. The Fibonacci construction is *anchored*: it begins at 1, 1, does not continue below as positive squares (its backward extension alternates in sign and passes through 0), and its ratios only *approach* φ – so while the whole figure may be resized, no arbitrary generation can be taken as the unit and regenerate the same series. (This contrast is with the *Fibonacci construction*, not with the golden spiral, which is itself a similarity; from the golden spiral we are separated by rate alone.)

Built side by side by the *same* rule – nested squares turned 45° at each step, which for Fibonacci is our comparison device, its standard construction turning 90° – the mechanism difference shows clearly. In the $\sqrt{2}$ spiral the 45° turn and the $\sqrt{2}$ scale are a single act: for square units, each square's diagonal *is* the next square's side, so nothing is imposed. A Fibonacci spiral drawn the same way must impose both separately – its sizes (1, 1, 2, 3, 5, 8, 13, ...) come from an additive recurrence, not from any diagonal, and a Fibonacci square's diagonal ($s\sqrt{2} \approx 1.414 s$) is not its successor ($\approx 1.6 s$). For square units, $\sqrt{2}$ is the unique ratio at which “the next is the diagonal” holds exactly (for a rectangle of ratio r the analogous ratio is $\sqrt{1 + r^2}$).

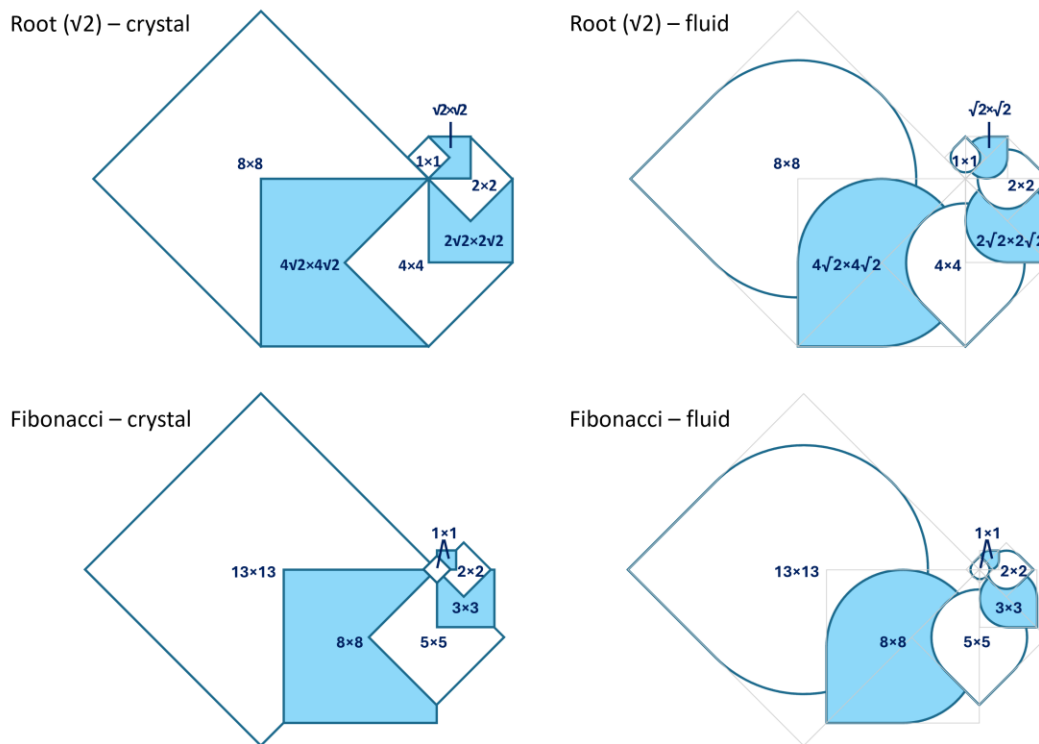


Figure 5. Apples to apples. Top: nested squares turned 45° at each step, sized by the $\sqrt{2}$ count (left: 1, $\sqrt{2}$, 2, ..., 8) and by the Fibonacci sequence (right: 1, 1, 2, 3, 5, 8, 13); the 45° turn is the paper's comparison device – the standard Fibonacci construction turns 90° . Bottom: the same frames rendered fluid – the teardrop glyph of §5, the analogue of the quarter-circle arcs of the familiar Fibonacci

picture. In the $\sqrt{2}$ spiral each square's diagonal is exactly the next square's side; in the Fibonacci built the same way, it is not.

The two can also be made to coincide *as curves* by tuning: a rectangle of ratio $\approx 1:1.535$ (turn $\approx 33.1^\circ$, about 10.9 generations per turn) reproduces golden per-quarter growth. But even then it remains this construction, not Fibonacci – a constant multiplicative ratio with no additive recurrence – and it forfeits the eight-fold parity. Curve and mechanism are separable; the claim “not Fibonacci” is about the mechanism, and holds regardless of tuning.

7. $\sqrt{2}$ as a language

Assembled, the construction reduces to a single object: the number $1 + i$, whose modulus is $\sqrt{2}$ and whose argument is 45° – the diagonal of the unit square, read at once as a length and as an angle. Everything in the figure is a power of it; the fixed pole (the uncounted origin) and the sense of turn complete the generator, and the alternating on/off-axis parity is not a separate ingredient but a consequence of the 45° .

What we wish to draw out is less a theorem than a stance. In this construction $\sqrt{2}$ is not a quantity computed and inserted; it is the medium. Position, scale, rotation, the two interleaved series, the crystal/fluid and bounded/aerated choices, and even the rectangular variants are all expressions of $\sqrt{2}$ – internal to the figure, never imported. Read this way, a collection of apparently separate features (a numberless count, a folded dimension, a shape-invariance, a not-quite-golden curve) resolves into one simple statement: $\sqrt{2}$ is the language the figure is written in, and the figure is what that language says when it counts by returning to zero.

This economy is generative. The same generator carries further constructions – other rational turns and growth ratios; the self-overlap structure that sets in from the third generation; the passage between the polygonal and the round that the crystal/fluid pair only begins to open – which the author develops elsewhere. Each is a further sentence in the same language.

Relation to known constructions

For honesty and for the reader's orientation, the classical antecedents used above, none claimed as original:

- **Doubling the square / the diagonal:** *Meno* 82b–85b; Vitruvius, *De architectura* IX pref. The square on the diagonal has twice the area.
- **$\sqrt{2}$ / dynamic rectangles / ad quadratum:** Hambidge's dynamic symmetry; the $\sqrt{2}$ (A-series) rectangle; the medieval *ad quadratum* construction (45° -rotated, $\sqrt{2}$ -scaled nested squares; Roriczer's 1486 booklet is the standard source).
- **The generator:** the Gaussian integer $1 + i$; multiplication by $1 + i$ as “scale $\sqrt{2}$, rotate 45° ”; the eight-fold closure as the order 8 of $(1 + i)/|1 + i|$, i.e. the cyclic rotation group C_8 (the rotation subgroup of the octagon's symmetry).

- **The curve:** the logarithmic / equiangular spiral – first described by Descartes (1638), rectified by Torricelli, and named *spira mirabilis* by Jacob Bernoulli; the spiral similarity (dilative rotation) in Coxeter and in Yaglom's *Geometric Transformations*.
- **The stroke:** a single-rule turtle spiral (turtle graphics: Papert); as a rewriting system it is an L-system (Lindenmayer, 1968), with the turtle interpretation of L-systems due to Prusinkiewicz.
- **A caution on names:** the $1:\sqrt{2}$ rectangle is sometimes called a “silver rectangle” (the Japanese *hakugin-hi* / silver-ratio convention), but “silver ratio” is also used for $1 : (1 + \sqrt{2})$; the growth here is $\sqrt{2}$, not $1 + \sqrt{2}$. Verify by the number, not the name.

Directions (developed further in the wider work)

- The self-overlap along the axes: it sets in from generation 3 (a generation over the previous-but-one's closing side, $k = 2$) and gains a second layer from generation 9 (containment of the body eight steps back, $k = 8$).
- Other rational turns and growth ratios on the same generator; when the eight-fold closure recurs.
- The polygonal $\leftrightarrow$ round passage (crystal/fluid) as a limit statement.
- The $1:\sqrt{2}$ “true-unit” spiral (turn $\arctan(1/\sqrt{2}) \approx 35.26^\circ$) as the sibling that trades the eight-fold parity for a different ratio.

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Supplementary materials

All companion files are in supplementary/.

- **Video S1** – the construction drawn as one continuous, no-lift stroke, in order, 32 edges (the moving form of Figure 1).
- **draw_root_spiral.py** – self-contained code (numpy + matplotlib) that draws the spiral from the generator $1 + i$, reproducing the geometry from first principles;
- **drawn_root_spiral.png** – the drawn spiral by the code (resulting figure).
- **Construction primer** – a step-by-step build of the stroke: the first unit square, its diagonal $\sqrt{2}$, and the square raised on that diagonal (a visual companion to §1–§2).
- **Figures** – the paper's five figures as high-resolution exports in figs_pub/, prepared from the author's presentation deck.